Efficient Hyperspectral Sparse Regression Unmixing With Multilayers

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***Abstract*— The sparse regression method is known for its ability to unmix hyperspectral data, but it can be computationally expensive and accurately insufficient due to the large scale and high coherence of the spectral library. To address this issue, a new approach called layered sparse unmixing termed LSU has been proposed in this article. This method involves breaking down the sparse unmixing process into multilayers, each of which interactively learns a row-sparsity-promoting abundance matrix and fine-tunes active library atoms based on measured activeness. By doing so, LSU outputs both a learned abundance matrix and an optimal library that can best model each mixed pixel in the scene. The proposed LSU can be efficiently solved by the alternating direction method of the multipliers framework. Experimental results obtained from simulated and real hyperspectral images demonstrate the effectiveness of LSU. The demo of the proposed LSU will be publicly available at https://github.com/XiangfeiShen/Layered_Sparse_Regression_Unmixing.**

***Index Terms*— Alternating direction method of multipliers (ADMM), hyperspectral analysis, hyperspectral unmixing, multiple layer, sparse regression, sparse unmixing.**

I. Introduction

HYPERSPECTRAL sparse regression unmixing amounts to finding a small number of pure spectral signatures in a (potentially very large) spectral library (also known as dictionary or endmember) that can best model each mixed pixel in the scene with sparse combination coefficients (also known as abundance fractions) [1], [2], [3], [4], [5]. Compared to the nonnegative matrix factorization-based blind unmixing [6], [7], [8], [9], [10], [11], [12], [13], [14] or endmember extraction-based least-square two-phase unmixing [15], [16], [17], [18], [19], [20], [21], [22], sparse regression unmixing methods tend to estimate the abundance matrix as sparsely as possible in a semisupervised behavior, where the library is assumed to be known in advance. This is because the number of endmembers present in a given scene is often small compared to the scale of the spectral library, and most importantly, the number of materials participating in a mixed pixel is usually on the order of 4–5 [3], [4].

The representative and systematic study on sparse regression unmixing was carried out by Bioucas-Dias and Figueiredo [23], which casts sparse regression unmixing in the form of a convex, constrained, sparse optimization problem [24]. In this study, the variable splitting and augmented Lagrangian framework is applied to solve the L_1-regularized sparse regression optimization problem. Inspired by [23], an $L_q(0 < q < 1)$-induced nonconvex sparse regularizer was discussed by Xu et al. [25], followed by some attempts that enhance the sparsity of the abundance matrix using an iterative reweighted strategy [26]. Wang et al. [27] introduced elementwise and channelwise iterative reweighted matrices to promote the sparsity of abundances. Similarly, Zhang et al. [28] extended the elementwise iterative reweighted approach by incorporating spatial contextual information. Several other studies have also applied the iterative reweighted strategy to the unmixing task, as documented in [29], [30], [31], [32], and [33]. Compared to L_1-based regularizer and its nonconvex extensions, Iordache et al. [34] hold the belief that only a few endmembers exist in the scene and the abundance matrix thus contains a low number of nonzero rows. This belief leads to the application of collaborative sparsity $L_{2,1}$ to the sparse regression model, which promotes the row sparsity of the abundance matrix. Also, several nonconvex versions of $L_{2,1}$ norm were exploited such as $L_{2,0}$ [35] and $L_{2,q}(0 < q < 1)$ [36]. Due to the smooth property of abundance maps in natural distribution of materials, spatially correlated pixels should share similar abundance values. In this regard, Iordache et al. [37] exploited the spatial contextual information present in the scene by introducing a total variation regularizer [38]. This method also served as inspiration for many other representative investigations such as nonlocal means-based sparse regression unmixing [39], nonlocal-weighted sparse regression unmixing [40], and convolution neural network-based sparse regression unmixing [41]. In hyperspectral images, some local regions with high homogeneity are normally mixed with a few substances, while transition areas contain more objects [42]. This observation triggers the discussions and applications of local sparsity. Zhang et al. [43] applied $L_{2,1}$

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to the preservation of local collaborativity. Borsoi et al. [44] explored local prior knowledge using superpixel techniques and minimized the errors between observed signatures and their corresponding local mean. Shen et al. [42] exploited a measurable local sparsity prior guided by superpixels to enhance unmixing performances.

The sparse regression unmixing algorithms mentioned above operate on the assumption that mixed pixels can be represented as a linear combination of a predefined set of pure spectral signatures, which is available in a library. For instance, the well-known one publicly available U.S. Geological Survey (USGS) spectral library[1] contains 1300 mineral signatures and the National Aeronautics and Space Administration Jet Propulstion Laboratory's Advanced Spaceborne Thermal Emission and Reflection Radiometer (NASA ASTER) spectral library[2] even has over 2400 spectra of natural and man-made materials. When spectral libraries are utilized to solve the unmixing problem, the process of estimating abundance no longer relies on pure spectral signatures being present in the input data or the ability of an endmember extraction algorithm to identify such signals but is obtained by searching optimal subset of atoms in the library that can best approximate mixed pixels in the scene [4]. However, the current spectral library-aided semisupervised sparse regression unmixing have the following flaws.

1) Existing investigations has been constrained by a single-layered analysis paradigm that relies on fixed inputs, thereby limiting their overall effectiveness. For a given spectral library $A \in \mathbb{R}^{B\times D}$ with B spectral bands and D library atoms, the single-layered sparse regression unmixing can be modeled by

$$\min_{X} \frac{1}{2}\|Y - AX\|_F^2 + \lambda\mathcal{R}(X) \quad \text{s.t. } X \succeq 0 \tag{1}$$

where Y and X denote the observed hyperspectral data and the estimated abundance matrix, respectively, $\|\cdot\|_F$ denotes the Frobenious norm, $\mathcal{R}(X) \equiv \|X\|_1$ stands for a sparsity regularizer, and λ denotes a penalty parameter. Since the abundance matrix X reflects the fractional distribution of the library, X subjects to a nonnegative constraint. When the library used to unmix (1) is large, it leads to a large percentage of inactive atoms in the library. However, such library cannot be fine-tuned based on the learned abundance matrix due to the single-layer unmixing behavior.

2) Spectral library, limited by its potentially high coherence and large scale, has hindered effective unmixing. To reduce the high coherence and correlation among atoms in the library, coupled regularizations (e.g., sparsity plus spatial smoothness [45], [46], [47], sparsity plus low rankness [33], [48], [49], [50], [51], [52], and sparsity plus graph Laplacian [53], [54], [55], [56]) are required to mitigate the unmixing errors. However, this process can exponentially increase computational resources and parameter tuning difficulty. To address these issues, several methods have emerged that fine-tune library atoms prior to the unmixing task [57], [58], [59]. One well-known dictionary fine-tuning approach is multiple signal classification (MUSIC) [60], which starts by estimating the signal subspace via the approach in [61] and then runs a binary test over the spectral library to identify active endmembers. This method inspires its robust extension proposed in [36] by introducing a vector to compensate for the error term. Nevertheless, these methods follow a data-driven manner, i.e., learn active atoms from the library based on the latent subspace of a given potentially contaminated hyperspectral data. If the data are severely contaminated with noise, these methods may mismatch active atoms in advance but propagate unmixing errors.

[1]Available online at http://speclab.cr.usgs.gov/spectral-lib.html
[2]Available online at http://speclib.jpl.nasa.gov

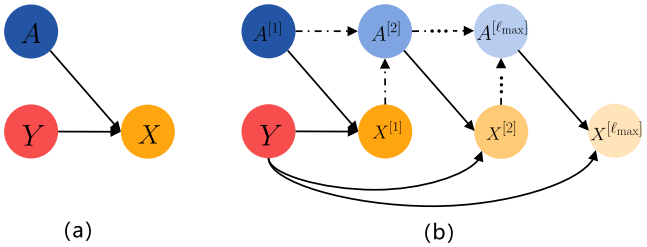


Fig. 1. Visual illustrations between two different unmixing paradigms. (a) Traditional sparse unmixing follows a single-layered paradigm that estimates the abundance X based on fixed inputs of hyperspectral data Y and library A. (b) Proposed LSU operates the unmixing task using mulilayers in which each layer estimates the abundance (e.g., $X^{[\ell]}$) according to fine-tuned library (e.g., $A^{[\ell]}$) and fine-tunes the library $A^{[\ell]}$ based on leaned $X^{[\ell]}$ for estimating the abundance $X^{[\ell+1]}$ at the next layer.

In this article, we introduce LSU, a layered sparse unmixing framework. Specifically, each layer of the LSU model involves estimating row-sparsity-promoting abundances based on a fine-tuned library within the inner loop. Building upon the previous study by Candes et al. [26], we enhance the sparsity of abundances through iterative reweighted weights. To optimize the inner loop, we employ the alternating direction method of multipliers (ADMM) framework. In the outer loop of the LSU model, we further refine the active library atoms based on their activeness, which is measured by the learned abundances. This iterative process allows us to select and fine-tune the most relevant library atoms layer-by-layer. By estimating abundances and selecting the appropriate library layer-by-layer, the LSU model automatically updates hyperparameters and finally terminates when the size of the spectral library matches the predefined threshold. Fig. 1 illustrates the comparisons between traditional and layered paradigms. Experimental results obtained from state-of-the-art techniques applied to simulated and real hyperspectral datasets validate the effectiveness and efficiency of our proposed method.

We make two chief contributions.

1) We propose breaking down the unmixing process into multilayers, each of which interactively learns a row-sparsity-promoting abundance matrix based on the iterative reweighted weights and fine-tunes active library atoms based on measured activeness. By iteratively estimating abundances and fine-tuning libraries

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layer-by-layer, LSU dynamically updates hyperparameters and ultimately terminates when the library size aligns with the predefined threshold. Compared the existing single-layered studies, our multilayer structure distinguishes itself through its novel combination and implementation.

2) We design an ADMM framework to solve the proposed optimization model. Experiments on simulated and real hyperspectral datasets indicate the effectiveness of LSU.

This article is divided into four sections, which can be given as follows. The LSU and its optimization framework are introduced in Section II in detail. Section III displays the results. Section IV concludes with some remarks.

II. Proposed Layered Sparse Regression Unmixing Model

The conventional approach to sparse regression unmixing involves fixed inputs, limiting its effectiveness. In this section, we introduce the proposed LSU method that extends single-layered unmixing to a multilayer paradigm by dynamically adjusting inputs based on learned information at each layer. To construct the multilayer structure, we focus on fine-tuning the library based on the learned row-sparsity-promoting abundances. This process comprises two parts. First, taking inspiration from the $L_{2,1}$ norm, we measure the activeness of learned abundances and sort them accordingly to fine-tune the library. Second, as the abundance's row sparsity indicates the importance of corresponding library atoms, we incorporate iterative reweighted weights into the L_1-induced sparsity to learn the row-sparsity-promoting abundance.

A. Learning to Fine-Tune Library via Activeness Measure

Due to the potentially large and highly correlated nature of the spectral library, sparse regression unmixing methods often require multiple regularizations to reduce its coherence so as to obtain sparse abundances. To address this issue, this article proposes a simple approach that fine-tunes libraries in a multiple layer fashion based on the learned abundance matrix. This approach offers two advantages: first, the fine-tuned libraries progressively approximate the true endmembers present in the data, and second, they alleviate the computational burden associated with traditional unmixing algorithms.

To fine-tune the library, we need to measure the zero or near-zero rows of the abundance matrix. In this regard, we propose an activeness measure. Suppose that $\mathcal{A} \in \mathbb{R}^{D\times 1}$ denotes a vector indicating the activeness of abundance rows. It can be computed based on the learned abundance matrix X by

$$\mathcal{A} = \sqrt{X \otimes X}\mathbf{1}_N \tag{2}$$

where $\otimes$ represents the elementwise multiplication operator and $\mathbf{1}_N$ denotes an all-one vector with dimension specified in its subscript.

By sorting $\mathcal{A}$ in descending order, the activeness of active endmembers in the library can be measured. Those endmembers with high activeness will be selected for subsequent unmixing tasks. By repeatedly operating the process mentioned above layer-by-layer until the size of the library matches the predefined p, we can gradually obtain the active atoms in the library that may be present in the scene and produce accurate abundances. Notably, due to the virtual dimentionality [62] and HySime [61] techniques, p can be estimated.

B. Learning to Promote Row Sparsity of Abundances

Due to the fact that the number of endmembers participating in a mixed pixel is usually very small compared with a spectral library, only a few of endmembers in the library are active, leading to the abundance matrix with row sparsity. This section considers applying rowwise and elementwise-based iterative reweighted approaches to the regularization of abundances, which is defined as

$$\mathcal{R}(X) \equiv \|W \otimes X\|_1 \tag{3}$$

where $W = W_1 \otimes W_2$ denotes the reweighted weight matrix, W_1 denotes the row-sparsity-promoting reweighted weight, and W_2 stands for the elementwise reweighted weight. W_1 can be defined as follows:

$$W_1 = 1 \oslash ((X \otimes X)\mathbf{1}_{N\times N} + \text{eps}) \tag{4}$$

where $\oslash$ represents the elementwise division, $\mathbf{1}_{N\times N}$ denotes an all ones matrix with N rows and N columns, and eps $>$ 0 denotes a stabilization parameter. For W_2, it is defined by

$$W_2 = 1 \oslash (|\mathcal{F}(X)| + \text{eps}) \tag{5}$$

where $\mathcal{F}(\circ)$ denotes a filter operation, which promotes the smoothness of reweighted weights.

C. Overview of the LSU Model

Suppose that $A^{[\ell]} \in \mathbb{R}^{B\times\delta^{[\ell]}}$ and $X^{[\ell]} \in \mathbb{R}^{\delta^{[\ell]}\times N}$ denote, respectively, the fine-tuned spectral library and corresponding coefficients at the ℓ layer, where $\delta^{[\ell]}$ represents the number of library atoms of $A^{[\ell]}$ at the ℓ layer. LSU can be formulated at ℓ layered by

$$\begin{aligned}\min_{X^{[\ell]}} \ & \frac{1}{2}\left\|Y - A^{[\ell]}X^{[\ell]}\right\|_F^2 + \lambda\left\|W^{[\ell]} \otimes X^{[\ell]}\right\|_1 \\ & + \frac{\rho^{[\ell]}}{2}\left\|\mathbf{1}_{\delta^{[\ell]}}^T X^{[\ell]} - \mathbf{1}_N^T\right\|_F^2 \\ \text{s.t. } & X^{[\ell]} \succeq 0\end{aligned} \tag{6}$$

where λ is a regularization parameter for the sparsity regularization, the first part maintains the minimum reconstruction error, the second term enforces the row sparsity by introducing the reweighted matrix W, and the third term improves the abundance sum-to-one (ASC) penalty by progressively increasing hyperparameter $\rho^{[\ell]}$. It should be noted that the ASC penalty is generally not considered due to the high mutual coherence of the library. However, when the number of the spectral library approximates to the real distribution of land covers, the ASC penalty can be imposed to produce optimal solutions [63].

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D. Optimization Framework

For simplicity, we only formulate the optimization procedures of LSU at the ℓ layer. We first reformulate (6) by the following constrained formulation:

$$\begin{aligned}\min_{X^{[\ell]},V_1^{[\ell]},V_2^{[\ell]},V_3^{[\ell]},V_4^{[\ell]}} & \frac{1}{2}\left\|Y-V_1^{[\ell]}\right\|_F^2+\lambda\left\|W^{[\ell]}\otimes V_2^{[\ell]}\right\|_1 \\ & +\frac{\rho^{[\ell]}}{2}\left\|\mathbf{1}_{\delta^{[\ell]}}^T V_3^{[\ell]}-\mathbf{1}_N^T\right\|_F^2+\iota_{\mathbb{R}_+}(V_4) \\ \text{s.t. } & V_1^{[\ell]}=A^{[\ell]}X^{[\ell]},\quad V_2^{[\ell]}=X^{[\ell]} \\ & V_3^{[\ell]}=X^{[\ell]},\quad V_4^{[\ell]}=X^{[\ell]}\end{aligned} \tag{7}$$

with its compact form expressed as

$$\min_{X,V} g(V),\quad \text{s.t. } GX+BV=0 \tag{8}$$

where $V=(V_1^{[\ell]},V_2^{[\ell]},V_3^{[\ell]},V_4^{[\ell]})$ denotes the auxiliary variable, $\iota_{\mathbb{R}_+}(\circ)$ denotes the nonnegative projector, $g(V)\equiv(1/2)\|Y-V_1^{[\ell]}\|_F^2+\lambda\|W^{[\ell]}\otimes V_2^{[\ell]}\|_1+(\rho^{[\ell]}/2)\|\mathbf{1}_{\delta^{[\ell]}}^T V_3^{[\ell]}-\mathbf{1}_N^T\|_F^2+\iota_{\mathbb{R}_+}(V_4)$, $G=[A;I;I;I]$, and $B=\mathrm{diag}(-I)$.

By introducing four scaled Lagrangian multipliers $D_1^{[\ell]}$, $D_2^{[\ell]}$, $D_3^{[\ell]}$, and $D_4^{[\ell]}$, the augmented Lagrangian form of (8) is expressed as

$$\begin{aligned}&\mathcal{L}_\mu\left(X^{[\ell]},V_1^{[\ell]},V_2^{[\ell]},V_3^{[\ell]},V_4^{[\ell]},D_1^{[\ell]},D_2^{[\ell]},D_3^{[\ell]},D_4^{[\ell]}\right) \\ &=\frac{1}{2}\left\|Y-V_1^{[\ell]}\right\|_F^2+\frac{\mu}{2}\left\|A^{[\ell]}X^{[\ell]}-V_1^{[\ell]}-D_1^{[\ell]}\right\|_F^2 \\ &\quad+\lambda\left\|W_1^{[\ell]}\otimes V_2^{[\ell]}\right\|_1+\frac{\mu}{2}\left\|X^{[\ell]}-V_2^{[\ell]}-D_2^{[\ell]}\right\|_F^2 \\ &\quad+\frac{\rho^{[\ell]}}{2}\left\|\mathbf{1}_{\delta^{[\ell]}}^T V_3^{[\ell]}-\mathbf{1}_N^T\right\|_F^2+\frac{\mu}{2}\left\|X^{[\ell]}-V_3^{[\ell]}-D_3^{[\ell]}\right\|_F^2 \\ &\quad+\iota_{\mathbb{R}_+}\left(V_4^{[\ell]}\right)+\frac{\mu}{2}\left\|X^{[\ell]}-V_4^{[\ell]}-D_4^{[\ell]}\right\|_F^2\end{aligned} \tag{9}$$

where $\mu>0$ denotes a penalty parameter. The ADMM iteratively minimizes $\mathcal{L}(X^{[\ell]},V^{[\ell]},D^{[\ell]})$ with respect to $X^{[\ell]}$, $V^{[\ell]}$, and $D^{[\ell]}$ in three consecutive steps until satisfying some stopping criterion, formulated as

$$\begin{cases}X^{[\ell,t+1]}\leftarrow\arg\min_X\mathcal{L}\left(X,V^{[\ell,t]},D^{[\ell,t]}\right)\\ V^{[\ell,t+1]}\leftarrow\arg\min_V\mathcal{L}\left(X^{[\ell,t+1]},V,D^{[\ell,t]}\right)\\ D^{[\ell,t+1]}\leftarrow D^{[\ell,t]}-GX^{[\ell,t+1]}-BV^{[\ell,t+1]}\end{cases} \tag{10}$$

where $X^{[\ell,t+1]}$ denotes the variable X at the ℓ layer and $(t+1)$th iteration. The detailed optimization details with respect to $X^{[\ell]}$, $V^{[\ell]}$, and $D^{[\ell]}$ can be found in the Appendix.

E. Pseudocode of LSU

The overall algorithmic pseudocode is displayed in Algorithm 1. The inner loop learns the abundance matrix according to the fine-tuned library, and the outer loop fine-tunes the libraries according to the activeness of learned abundance matrix. By interactively executing two parts, we obtain the final abundance matrix and fine-tuned library that can best model each mixed pixel in the scene. It is worth mentioning that during the library fine-tuning process, $1/q$ of library atoms will be removed.

Algorithm 1 Pseudocode of LSU

Require: $Y\in\mathbb{R}^{B\times N}$, $A^{[1]}\in\mathbb{R}^{B\times\delta^{[1]}}$, λ, p, q, $\rho^{[1]}$, $\tau^{[1]}$;
Ensure: X

while $\delta^{[\ell]}!=p$ **do**
  Initialize $V_1^{[\ell]}=0$, $V_2^{[\ell]}=0$, $V_3^{[\ell]}=0$, $V_4^{[\ell]}=0$, $D_1^{[\ell]}=0$, $D_2^{[\ell]}=0$, $D_3^{[\ell]}=0$, $D_4^{[\ell]}=0$.
  repeat
    % Inner Loop for Estimating X %
    Compute $X^{[\ell]}$ via formula (18);
    Compute $V_1^{[\ell]}$ via formula (20);
    Compute $V_2^{[\ell]}$ via formula (22);
    Compute $V_3^{[\ell]}$ via formula (24);
    Compute $V_4^{[\ell]}$ via formula (26);
    Compute $D_1^{[\ell]}$, $D_2^{[\ell]}$, $D_3^{[\ell]}$, $D_4^{[\ell]}$ via formula (27);
    Update inner iterations;
  until Stopping criteria are satisfied
  % Outer Loop for Fine-tuning A %
  Obtain $\mathcal{A}$ by computing (2);
  Sort $\mathcal{A}$ in ascending order and get the indices $\mathcal{I}$;
  if $\delta^{[\ell]}==p$ **then**
    Break the loop;
  else if $\delta^{[\ell]}-\delta^{[\ell]}/q\le p$ **then**
    Update library $A^{[\ell]}$ by $A^{[\ell]}(:,\mathcal{I}(1:\delta^{[\ell]}-p))=[]$
    Update library size by $\delta^{[\ell]}=p$;
  else if $\delta^{[\ell]}-\delta^{[\ell]}/q>p$ **then**
    Update $A^{[\ell]}$ by $A^{[\ell]}(:,\mathcal{I}(1:\lceil\delta^{[\ell]}/q\rceil))=[]$
    Update library size by $\delta^{[\ell]}=\delta^{[\ell]}-\lceil\delta^{[\ell]}/q\rceil$;
  end if
  Update ℓ by $\ell\leftarrow\ell+1$;
  % Update Hyperparameters %
  Update $\rho^{[\ell]}$ by $\rho^{[\ell]}\leftarrow\min(2*\rho^{[\ell]},\rho_{\max})$;
  Update $\tau^{[\ell]}$ by $\tau^{[\ell]}\leftarrow\min(2*\tau^{[\ell]},\tau_{\max})$;
end while

F. Adjustment of Hyperparameters

Since the size of library A goes down by progressively fine-tuning $1/q$ of atoms itself, the library A gradually decreases the coherence and redundancy of itself and tends to be the true endmember that present in the scene. Hence, the adjustment of hyperparameters follows a coarse-to-fine tuning process. Concretely, for the maximum iteration (named τ for readability), it is initially fixed at a small real number, e.g., $\tau^{[1]}=10$, and then updated by $\tau^{[\ell]}\leftarrow 2*\tau^{[\ell]}$ until subjected to the constraint $\tau_{\max}$, e.g., $\tau_{\max}=200$. Similarly, ρ is set to a real small, e.g., $\rho^{[1]}=0.1$, at the beginning, and then updated by $\rho^{[\ell]}\leftarrow 2*\rho^{[\ell]}$ until subjected to the constraint $\rho_{\max}$, e.g., $\rho_{\max}=15$ [12], [13]. To sum up, when the library A highly approximates the true signatures lying in the observed hyperspectral data, the learned abundance matrix should highly follow an ASC constraint and require sufficient iterations.

For the selection of q, it involves how many atoms will be removed and determines the number of layers. Let $\ell_{\max}$ denote the number of layers. As $\delta^{[\ell]}$ is progressively decreased, $\ell_{\max}$

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can merely be approximated by

$$\ell_{\max} \approx \text{round}\left(1 + \log_{\frac{q-1}{q}} \frac{p}{\delta^{[1]}}\right) \tag{11}$$

where round$(\circ)$ rounds each element of $\circ$ to the nearest integer.

G. Computational Complexity Analysis

For the ℓ layer, the abovementioned ADMM optimization framework involves successively updating $X^{[\ell]}$, $V_1^{[\ell]}$, $V_2^{[\ell]}$, $V_3^{[\ell]}$, $V_4^{[\ell]}$, $D_1^{[\ell]}$, $D_1^{[\ell]}$, $D_3^{[\ell]}$, and $D_4^{[\ell]}$. Since F in (18) can be precalculated, the computational complexity of (18) is $\mathcal{O}((\delta^{[\ell]})^2 N + \delta^{[\ell]} NB)$. The computational complexity of (20) is $\mathcal{O}(\delta^{[\ell]} NB)$. The computational complexity of (22), (26), and (27) is $\mathcal{O}(\delta^{[\ell]} N)$. Similar to (18), the calculation of inverse matrix of (24) can be precalculated, the computational complexity is $\mathcal{O}((\delta^{[\ell]})^2 B)$. Overall, the computational complexity of LSU at the ℓ layer is $\mathcal{O}((\delta^{[\ell]})^2 N + (\delta^{[\ell]})^2 B + \delta^{[\ell]} NB + \delta^{[\ell]} N)$. Due to the library fine-tuning process in each layer, δ drops increasingly until matches the predefined p. Therefore, the computational complexity of the final layer is $\mathcal{O}(p^2 N + p^2 B + pNB + pN)$.

III. Experimental Results and Analysis

A. Experimental Setup

1) Benchmark Methods: To fairly compare the proposed LSU with other competitors, 11 state-of-the-arts are considered, including sparse unmixing via variable splitting and augmented Lagrangian (SUnSAL) [23],[3] SUnSAL with total variation (SUnSAL-TV) [37],[4] collaborative sparsity version of SUnSAL (CLSUnSAL) [34],[5] double reweighted sparse unmixing with total variation (DRSU-TV) [27], spatial–spectral weighted sparse unmixing (S^{2}WSU) [28], nonnegative matrix factorization-minimum quadratic variation (NMF-MQV) [64],[6] multiscale sparse unmixing algorithm (MUA) [44],[7] two-step iterative-based weighted collaborative sparse unmixing with total variation (TSWCSU-TV) [45],[8] sparse unmixing with convolution neural network (SUnCNN) [41],[9] spatial–spectral low-rank constrained sparse unmixing (SSLRSU) [52], and local–global-based sparse unmixing (LGSU) [42].[10] Despite the SUnCNN conducted on the PyTorch, other methods are implemented by the MATLAB 2016a platform and run on a PC with an Intel Core i7-10700 and 16-GB RAM.

2) Evaluation Metrics: Two experimental validation metrics are used to verify the effectiveness of unmixing algorithms. Suppose that X denotes the true abundance and $\hat{X}$ stands for the estimated abundance. The first metric is called signal reconstruction error (SRE), which is defined as

$$\text{SRE(dB)} = 10 \times \log_{10}\left[\frac{E\left(\|X\|_F^2\right)}{E\left(\left\|X - \hat{X}\right\|_F^2\right)}\right] \tag{12}$$

where $E(\circ)$ denotes the expectation function. The higher the SRE values, the better the unmixing performances.

The second metric is root-mean-square error (RMSE) that measures the error between the measured and true values, which is given by

$$\text{RMSE} = \sqrt{\frac{1}{D \times N}\left\|X - \hat{X}\right\|_F^2}. \tag{13}$$

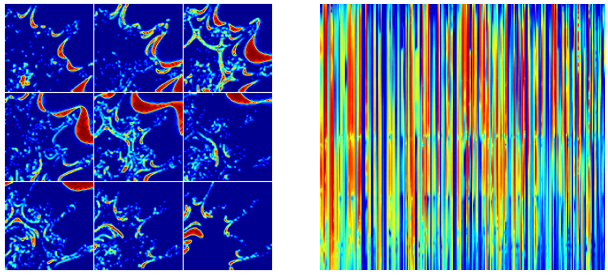
Low RMSE values indicate better unmixing performances.

For the Cuprite dataset, it does not have true values, implying that (13) cannot be applied to measure the abundance estimation directly. Therefore, we propose to measure the reconstruction error of Cupritie signatures based on the given library and estimated abundance, which is defined as

$$\text{RMSE}_{\text{Cup}} = \sqrt{\frac{1}{B \times N}\left\|Y - A\hat{X}\right\|_F^2}. \tag{14}$$

Low RMSE_{Cup} values indicate better unmixing performances.

3) Experimental Datasets: To provide quantitative results, we consider three datasets in our study. The first dataset (referred to as DC1) is generated based on a real-world fractal scenario. The ground truth of abundances of DC1 indicates that the pixels at the border of the region are heavily mixed, while those at the center are spectrally pure. DC1 comprises 100×100 pixels and is mixed by nine endmembers with nine abundance maps satisfying ASC and nonnegativity constraints. Nine spectral endmembers used to construct DC1 are selected from the USGS library that contains 188 bands and 498 library atoms. The spectral library used to unmix the DC1 derives from the USGS library with the angle between any pair of spectral signatures being at least 4.44°. In this regard, the spectral library contains 224 bands and 240 spectral atoms. The Gaussian noise is included in DC1 for three levels of signal-to-noise ratio (SNR): 30, 40, and 50 dB. Fig. 2 visually shows the abundance map and library of the DC1 dataset.



(a) (b)

Fig. 2. True abundance maps and library of the DC1 dataset. (a) True abundance map. (b) Library.

The second dataset (DC2) is generated based on the Dirichlet distribution centered at a Gaussian random field with each abundance map being piecewise smooth with sharp transitions in the border of the region generated. We select nine endmembers from the NASA Johnson Space Center (JSC)

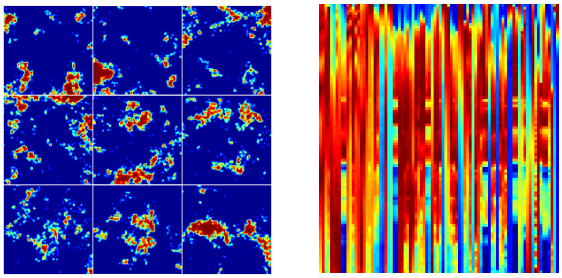
[3]Available online at http://www.lx.it.pt/~bioucas/code.htm
[4]Available online at http://www.lx.it.pt/~bioucas/code.htm
[5]Available online at http://www.lx.it.pt/~bioucas/code.htm
[6]Available online at https://github.com/LinaZhuang/NMF-QMV_demo
[7]Available online at https://github.com/ricardoborsoi/MUA_SparseUnmixing
[8]Available online at: https://github.com/jinju123/TSWCSU-TV
[9]Available online at https://github.com/BehnoodRasti/SUnCNN
[10]Available online at https://github.com/XiangfeiShen/2022_IEEE_GRSL_LGSU

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TABLE I

SRE AND RMSE RESULTS OBTAINED FROM ALL SPARSE UNMIXING ALGORITHMS ON DC1 AND DC2 DATASETS UNDER DIFFERENT NOISE LEVELS. THE BEST RESULTS ARE IN BOLD

Algorithm	Venue	DC1						DC2					
		30dB		40dB		50dB		30dB		40dB		50dB	
		SRE↑	RMSE↓	SRE↑	RMSE↓	SRE↑	RMSE↓	SRE↑	RMSE↓	SRE↑	RMSE↓	SRE↑	RMSE↓
SUnSAL [23]	**WHISPER-10**	7.6631	0.1012	13.9127	0.0493	19.4341	0.0261	10.2027	0.0817	15.9125	0.0424	20.7220	0.0244
SUnSAL-TV [37]	**TGRS-12**	9.9250	0.0780	16.4421	0.0368	22.8900	0.0175	12.0649	0.0660	19.1304	0.0292	24.4856	0.0158
CLSUnSAL [34]	**TGRS-14**	4.2918	0.1492	10.2214	0.0754	17.5071	0.0326	6.9998	0.1182	13.7781	0.0542	21.1607	0.0232
DRSU-TV [27]	**GRSL-17**	13.3038	0.0529	24.8901	0.0139	35.6271	0.0040	15.1163	0.0464	25.8970	0.0134	35.5321	0.0044
S^2WSU [28]	**TGRS-18**	13.6870	0.0506	25.8294	0.0125	35.9231	0.0039	15.8048	0.0429	26.2323	0.0129	37.1753	0.0037
NMF-QMV [64]	**TGRS-19**	8.2047	0.0951	15.1120	0.0429	23.8570	0.0157	6.1874	0.1298	12.3801	0.0636	21.0464	0.0235
MUA [44]	**GRSL-19**	8.5216	0.0917	10.2128	0.0754	10.7949	0.0706	9.6941	0.0867	11.0476	0.0742	11.2894	0.0721
TSWCSU-TV [45]	**AMC-21**	12.9541	0.0550	24.7535	0.0141	32.1676	0.0060	12.3724	0.0637	24.1438	0.0164	33.2953	0.0057
SUnCNN [41]	**GRSL-21**	17.1723	0.0339	23.2181	0.0169	27.3455	0.0105	14.4823	0.0499	24.4022	0.0159	30.9387	0.0075
SSLRSU [52]	**JSTARS-21**	14.2824	0.0472	27.3894	0.0104	39.0336	0.0027	11.4953	0.0704	26.5688	0.0124	36.6749	0.0039
LGSU [42]	**GRSL-22**	17.6179	0.0322	28.0765	0.0096	37.4133	0.0033	18.6917	0.0308	28.0501	0.0105	37.4219	0.0036
LSU	**/**	**20.6055**	**0.0228**	**30.6145**	**0.0072**	**40.7609**	**0.0022**	**19.7377**	**0.0273**	**29.2831**	**0.0091**	**38.7428**	**0.0031**



(a) (b)

Fig. 3. True abundance maps and library of the DC2 dataset. (a) True abundance map. (b) Library.

library, which contains 262 spectral signatures and 100 bands. In addition, the spectral library used to unmix DC2 is selected from the NASA JSC library with the angle between any pair of spectral signatures being at least 4.44°. After the pruning process, the library contains 88 atoms and 99 bands [65]. The Gaussian noise is included in DC2 for three levels of SNR: 30, 40, and 50 dB. Fig. 3 visually shows the abundance map and library of the DC2 dataset.

The third dataset is Cuprite captured by the airborne visible infrared imaging spectrometer (AVIRIS) sensor with its subimage being considered. The subimage of Cuprite contains 250 × 191 pixels and 188 pruned spectral bands. Fig. 4 shows the mineral distribution of the Cuprite dataset using a false color image, with the red rectangle indicating the location used in the experiment.[11] It is worth mentioning that the public AVIRIS Cuprite dataset was gathered in 1997, while the mineral distribution map was created in 1995. Therefore, it may not be entirely appropriate to make a direct comparison between them. Nevertheless, given the absence of available real abundance maps, the Tricorder map can serve as a point of reference for a qualitative analysis of the abundance maps generated by the unmixing methods under consideration. More importantly, we use the USGS library to unmix Cuprite data.

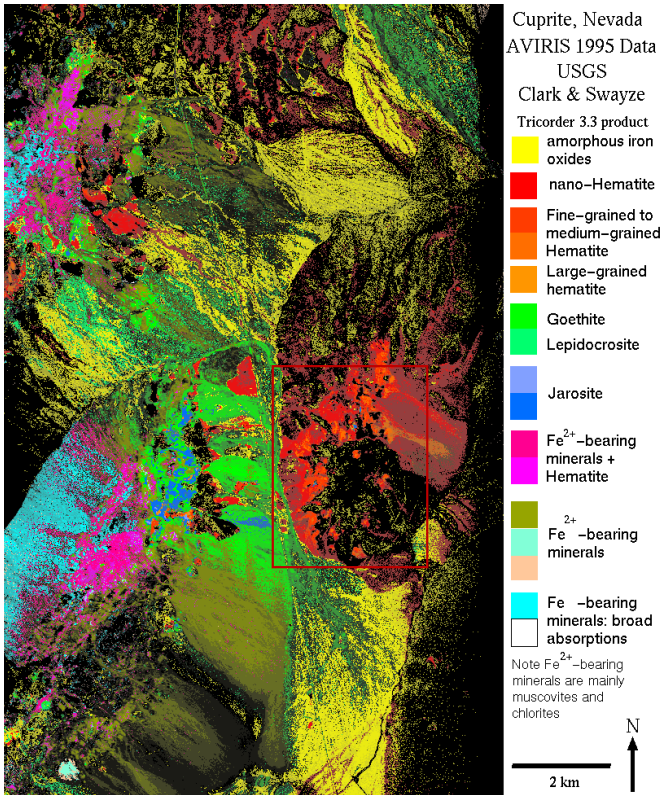


Fig. 4. USGS map showing the location of different minerals in the Cuprite mining district in NV.

B. Experiments on Simulated and Real Datasets

1) Experimental Results Obtained From Simulated Datasets: The experimental results of our proposed method and its competitors on two datasets are tabulated in Table I, with the best outcomes highlighted in bold. Our proposed method fixes p at 9 based on the estimation result of HySime on both datasets. As shown in Table I, our proposed method achieves superior SRE and RMSE results across all SNR levels and both datasets. Among the competitors, LGSU exhibits the second-best performance in terms of SRE and RMSE on DC1 and DC2 at all SNRs except for SSLRSU on DC1 under 50 dB.

To provide a visual representation of the abundance maps estimated by 12 unmixing algorithms and their corresponding GT, Figs. 5 and 6 have been included in which the colorbar of each type of dataset is listed in the bottom. For two datasets, CLSUnSAL produces poor estimation performances as it heavily emphasizes row sparsity. The visual effect of abundance estimation in NMF-QMV is hindered by the dual optimization variables of the endmember and abundance. Two

[11] Available online at https://www.usgs.gov/media/images

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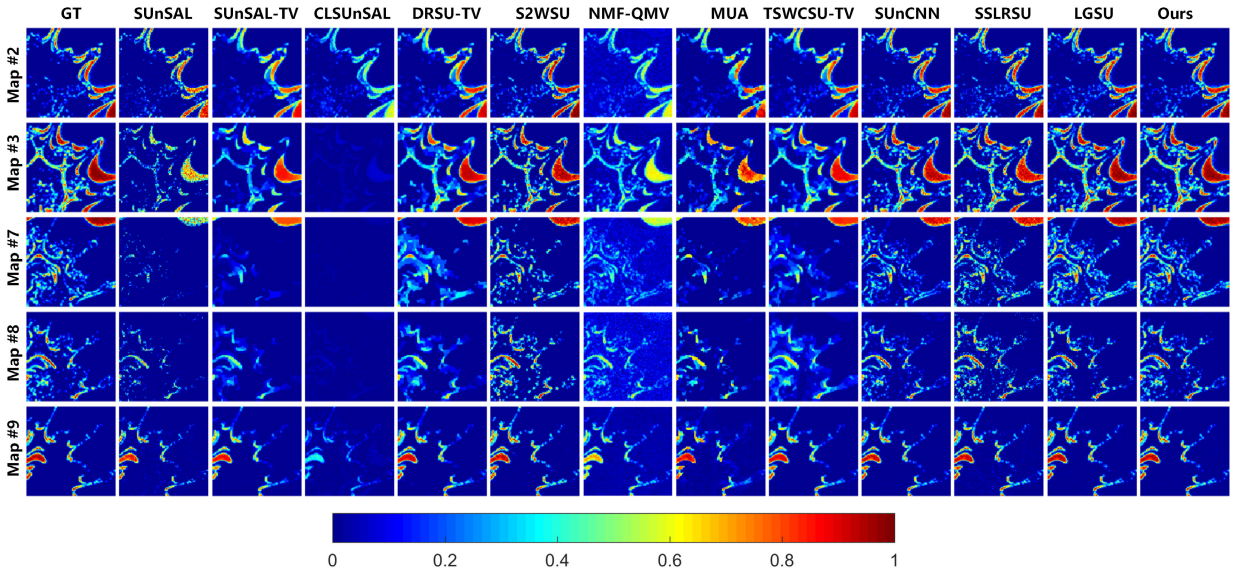


Fig. 5. Visual comparisons among the four representative maps derived from 12 algorithms and ground truths on the DC1 dataset with 30-dB noise.

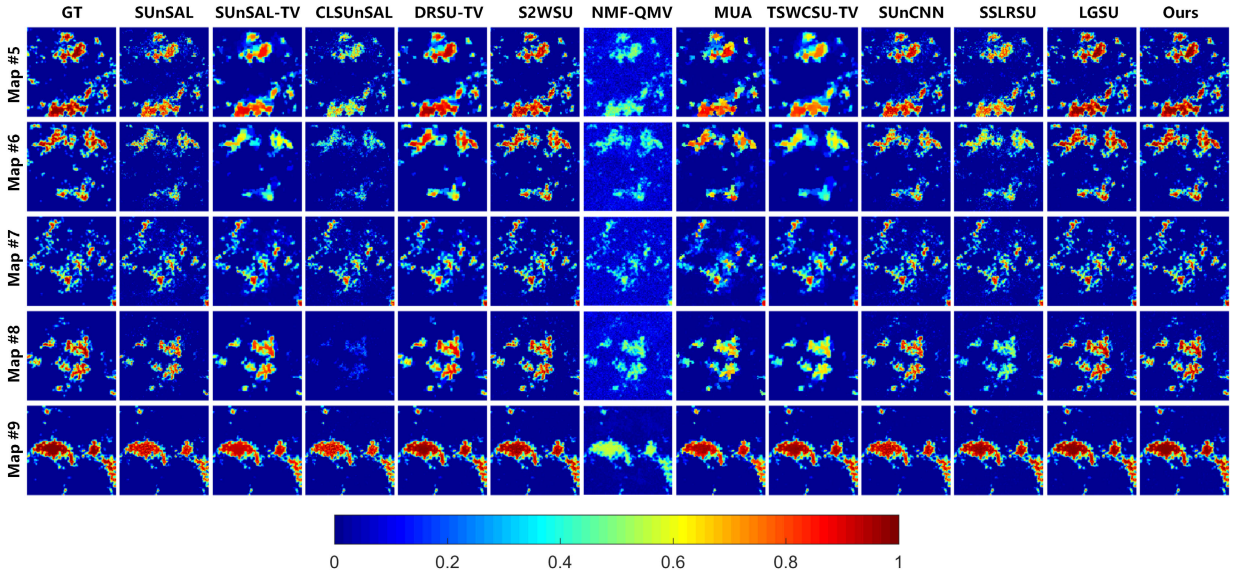


Fig. 6. Visual comparisons among the four representative maps derived from 12 algorithms and ground truths on the DC2 dataset with 30-dB noise.

superpixel-guided sparse unmixing algorithms, namely, MUA and LGSU, exhibit distinct estimation performances. Comparatively, MUA places greater emphasis on the correlation between the observed signature and its corresponding local mean, as opposed to LGSU that focuses highly on the sparsity of local areas. Total variation-regularized unmixing algorithms, such as SUnSAL-TV, DRSU-TV, and TSWCSU-TV, present an oversmoothed visual effect. These degradation phenomena exist in Fig. 6, but they are not significant in the abundance maps estimated by LGSU and our proposed method.

2) Experimental Results Obtained From the Cuprite Dataset: For the Cuprite dataset, the proposed method fixes $\tau^{[1]}$ at 30 and sets p to 30 due to the complex topological distribution of the Cuprite and the high coherence of the spectral library. The number of endmembers, which is fixed at 12 based on HySime, is a crucial hyperparameter for NMF-QMV. In accordance with previous studies [23], [34], [37], five materials (Alunite, Buddingtonite, Chalcedony, Kaolin/Smect, and Montmorillonite) are prominent in the Cuprite area. The resulting mineral abundance maps estimated by 12 unmixing methods and provided by the Tricorder software[12] are shown in Fig. 7, where the colorbar of each type of mineral abundance maps is listed in the rightmost. Analysis of these maps reveals four key findings. First, the proposed method produces desirable abundance maps without an oversmooth visual effect that outperforms those generated by other methods. The regularization problem causes the NMF-QMV abundance maps to not be sparse enough. Second, the proposed method requires only 279.5 s in total, making it one of the fastest methods available. Third, even though

[12]Available online at http://speclab.cr.usgs.gov/PAPER/tetracorder

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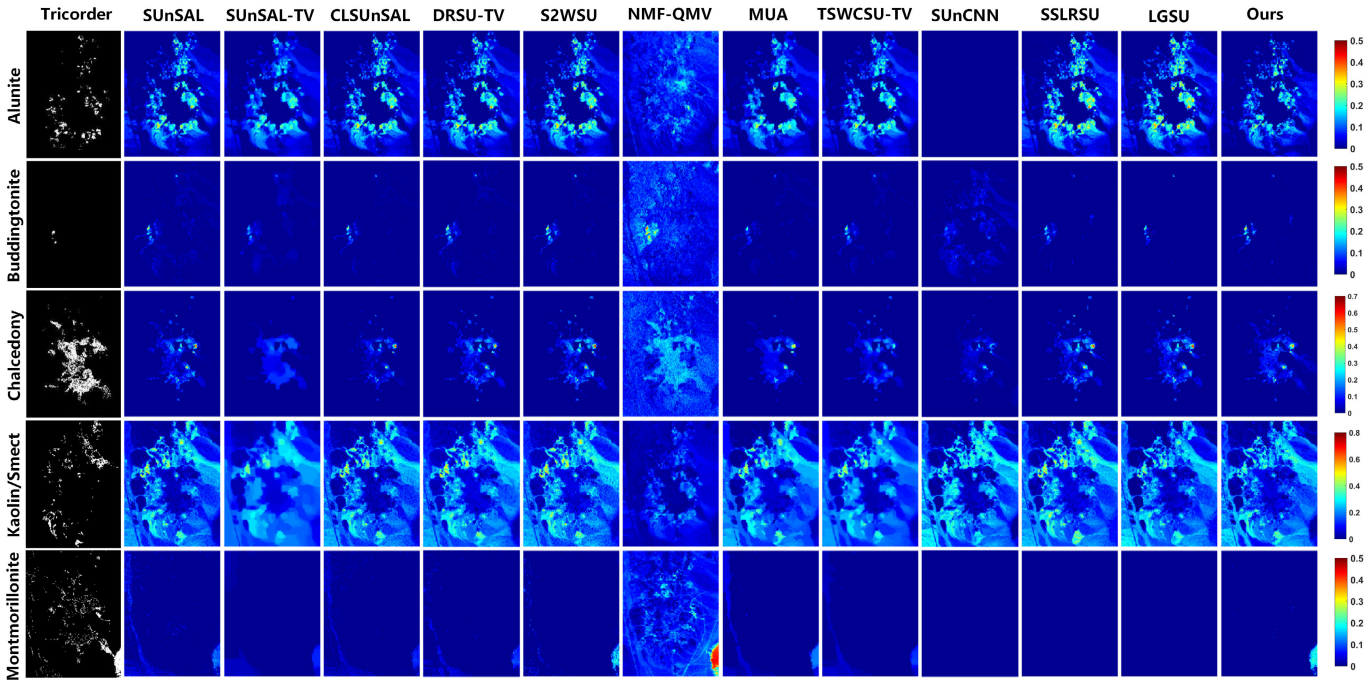


Fig. 7. Visual comparisons among the three representative maps derived from 12 algorithms and tricorder on the Cuprite dataset.

TABLE II

COMPARISON OF $RMSE_{Cup}$, RUNNING TIME (IN SECONDS), ITERATION, AND NO. OF LIBRARY ON THE CUPRITE DATA

Algorithm		$RMSE_{Cup}\downarrow$	Time↓	Iteration	No. of Library Atom
SUnSAL		0.0402	108.5	200	498
SUnSAL-TV		0.0054	854.1	200	498
CLSUnSAL		0.0053	101.8	200	498
DRSU-TV		0.0050	936.4	200	498
S^2WSU		0.0051	320.9	200	498
NMF-QMV		0.2330	1263.2	200	12
MUA		0.0248	82.72	200	498
TSWCSU-TV		0.0052	912.7	200	498
SUnCNN		0.0064	677.15	200	498
SSLRSU		0.0056	1011.5	200	498
LGSU		0.0061	806.8	200	498
LSU	Layer 1	0.0051	60.0	30	498
	Layer 2	0.0051	55.1	60	249
	Layer 3	0.0056	54.0	120	125
	Layer 4	0.0062	50.3	200	63
	Layer 5	0.0064	30.5	200	32
	Layer 6	0.0064	29.6	200	30
	Total	/	279.5	810	/

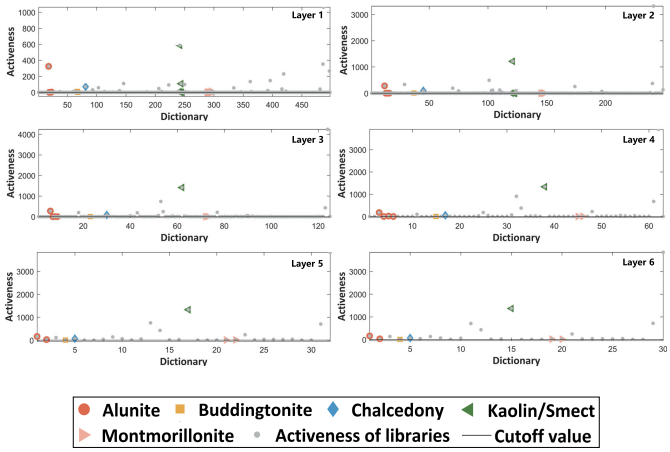


Fig. 8. Library fine-tuning details under different layers on the Cuprite data.

the proposed method reduces the size of the library by 94%, it still preserves the spectra associated with the most prominent materials. Fourth, the $RMSE_{Cup}$ values obtained from the proposed method are comparable to those obtained by our competitors. Specifically, in the first two layers, the $RMSE_{Cup}$ values derived from our method are lower than 0.0051.

Table II provides insight into the unmixing details of each layer. The $RMSE_{Cup}$ values derived from different layers of the proposed method are comparable to those from competitors, except for SUnSAL. However, as the proposed method progresses through each layer, it approaches the actual distribution of endmembers present in Cuprite. Fig. 8 visually shows this fine-tuning process, where inactive libraries are removed, but active libraries are well preserved.

3) Computational Complexity Analysis on Simulated and Real Datasets: In addition to the unmixing results presented in Table I, we conducted a thorough analysis of the running time (in seconds) for all algorithms on three different datasets, which are provided in Tables II and III. Our findings illustrate that our proposed method has the third lowest running time compared to other advanced approaches across all three datasets. For example, LSU requires 810 iterations to produce final abundances but only requires 279.5 s. This is obvious because state-of-the-art methods, such as SUnCNN, LGSU, SSLRSU, and TSWCSU-TV, consume many computational resources due not only to the large scale of the library applied in the single-layered analysis fashion but to the coupled regularization structures. By considering the results from

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TABLE III
RUNNING TIME (IN SECONDS) OBTAINED FROM ALL SPARSE UNMIXING ALGORITHMS ON DC1 AND DC2

Dataset	SUnSAL	SUnSAL-TV	CLSUnSAL	DRSU-TV	S^2WSU	NMF-QMV	MUA	TSWCSU-TV	SUnCNN	SSLRSU	LGSU	Ours
DC1	2.3	113.5	8.1	120.1	39.2	49.5	6.7	114.1	120.0	248.6	114.3	24.2
DC2	1.4	41.2	2.3	46.2	12.9	55.1	3.8	45.5	80.9	78.4	63.3	8.2

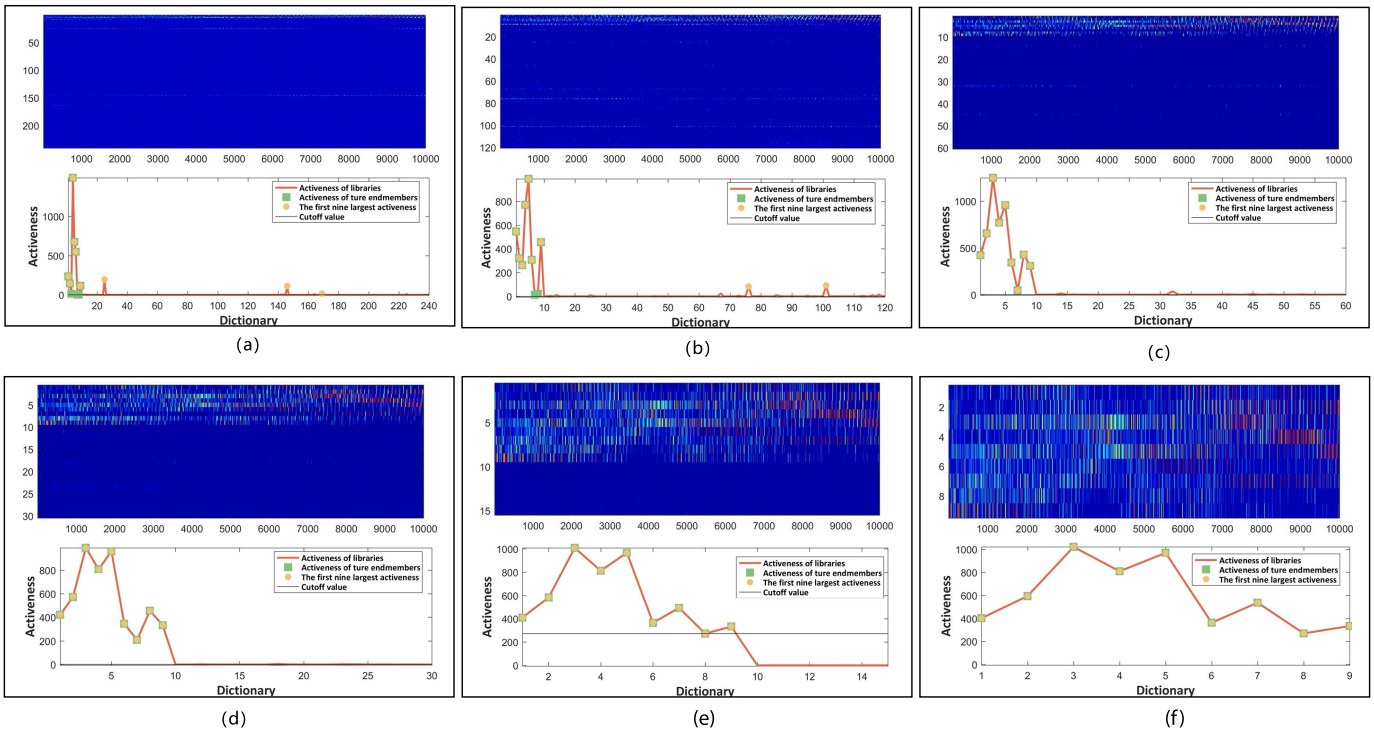


Fig. 9. Visualization of the abundance matrix and activeness measure at each layer on DC1 with 30-dB noise. (a) First layer. (b) Second layer. (c) Third layer. (d) Fourth layer. (e) Fifth layer. (f) Sixth (final) layer.

Tables I–III, we can confidently conclude that our proposed method is both effective and efficient.

C. Analysis of Layered Unmixing Structure

1) Visualization of Different Layer Details: To observe the unmixing details of each layer in the proposed method, this experiment was conducted based on DC1 with 30-dB noise. As seen from Fig. 9, in the first and second layers, there is a large percentage of the nonactive rows of the abundance matrix despite the scenario with the limited number of iterations. From the third layer onward, the nine largest activeness values corresponded to the indices of endmember referees. In the final layer, the proposed algorithm was able to output the learned abundances because the number of abundance rows matched the predefined p.

Fig. 10 shows the SRE values at each layer regarding iterations on DC1 with 30-dB noise. It can be seen from Fig. 10(a) that due to the large library scale, the proposed method uses small iterations (e.g., only ten iterations in the first layer) in initial layers to obtain the coarse abundances and reduce computational resources. As the library scale decreases, iteration numbers increase for improving unmixing performances. In the fifth layer, the proposed method achieves the highest SRE based on 15 library atoms within 100 iterations,

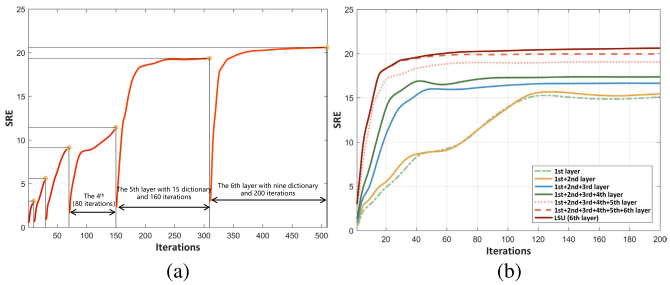


Fig. 10. Visualization of the SRE regarding iterations at each layer on DC1 with 30-dB noise. (a) Unmixing behavior of each layer of LSU. (b) Unmixing behavior comparisons between the final layer of LSU and different layer combinations.

while in the final layer, it extensively attains the highest SRE results based on only nine library atoms within 150 iterations.

2) Visualization of Different Layer Combinations: Fig. 10(b) visually compares the unmixing behavior of different layer combinations and the final layer of LSU. The quantitative SRE results of Fig. 10(b) can be observed from Table IV that tabulates the unmixing results on two kinds of cases. The first case unmixes hyperspectral data based on different layer combinations with 200 iterations and fixed ASC penalty. The second follows a coarse-to-fine paradigm that progressively increases the number of iterations and the penalty of ASC. Due to sufficient iterations, the unmixing

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TABLE IV

SRE, RMSE, AND TIME (IN SECONDS) RESULTS OBTAINED FROM TWO TYPES OF CASES ON DC1 WITH 30-DB NOISE

Scenario		SRE↑	RMSE↓	Time↓	Iterations	No. of Library Atom
Case 1	Layer 1	15.1010	0.0118	53.2	200	240
	Layer 1 + Layer 2	15.5425	0.0158	22.5	200	120
	Layer 1 + Layer 2 + Layer 3	16.8313	0.0193	12.6	200	60
	Layer 1 + Layer 2 + Layer 3 + Layer 4	17.5787	0.0250	8.1	200	30
	Layer 1 + Layer 2 + Layer 3 + Layer 4 + Layer 5	19.2437	0.0267	5.8	200	15
	Layer 1 + Layer 2 + Layer 3 + Layer 4 + Layer 5 + Layer 6	19.9532	0.0246	4.9	200	9
Case 2(Paradigm of LSU)	Layer 1	3.0260	0.0473	2.6	10	240
	Layer 2	5.6231	0.0496	2.4	20	120
	Layer 3	9.1523	0.0467	2.6	40	60
	Layer 4	11.5114	0.0503	3.2	80	30
	Layer 5	19.5536	0.0257	4.6	160	15
	Layer 6 (Output layer)	20.6055	0.0228	4.9	200	9
	Total	/	/	20.3	510	/

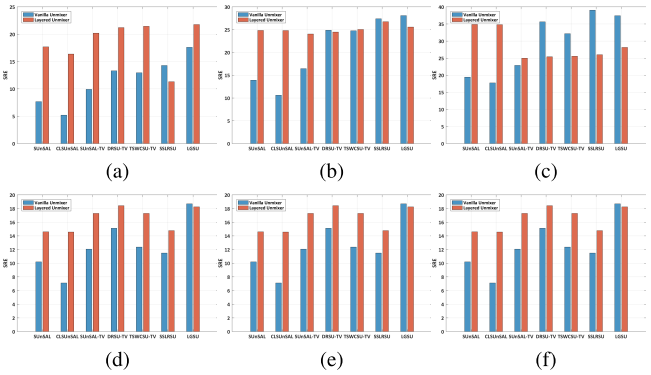


Fig. 11. Performance comparisons between the vanilla unmixer and its corresponding layered version. (a) DC1 (30 dB). (b) DC1 (40 dB). (c) DC1 (50 dB). (d) DC1 (30 dB). (e) DC2 (40 dB). (f) DC2 (50 dB).

performances of the first case are relatively high compared to those of the second. However, the unmixing performance of the output layer of the second case is much better than that of the first case. In addition, the second case requires only 20.3 s. These observations indicate that the proposed LSU can efficiently help boost the unmixing performance.

3) Can the Proposed Layered Structure Improve Vanilla Unmixer?: Given the significant improvements of the proposed method over seven competitors, a natural question arises: can the proposed layered structure improve vanilla unmixer? To answer the question, this experiment implemented seven competitors in the layered paradigm in which the original sparse regression method is called vanilla unmixer for short and that being layeredly modified is referred to as layered unmixer. These unmixers were conducted on DC1 and DC2 under three noise levels. Besides, all layered unmixers fine-tune half of the libraries without considering the ASC penalty.

Fig. 11 visually compares the vanilla unmixer and its corresponding layered version. Our observations led us to three conclusions. First, for the DC2 dataset, layered unmixers produce higher SRE values compared to the vanilla unmixers, except for the LGSU. Second, for DC1, due to the heavy fine-tuning level, most layered unmixers experience a mismatch issue, resulting in low SRE values in most cases. Third, the layered unmixers induced by SUnSAL and CLSUnSAL show clear improvements in unmixing performance compared to their vanilla versions. Therefore, we conclude that the layered

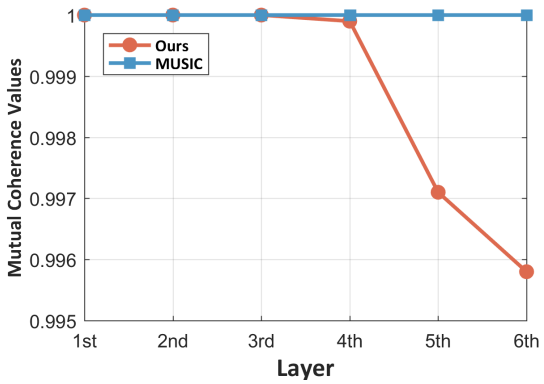


Fig. 12. Mutual coherence comparisons between the MUSIC and the proposed layered structure and its corresponding layered version.

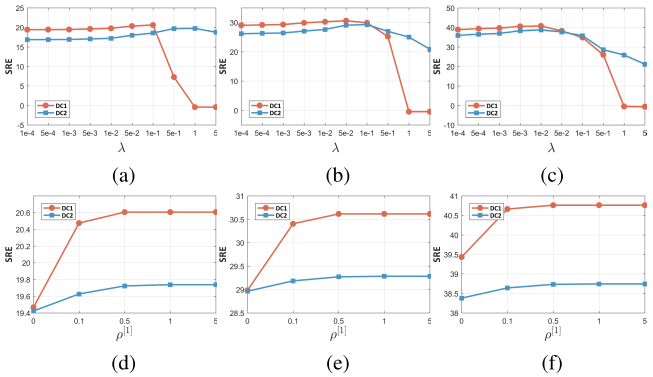


Fig. 13. Parameter tuning regarding λ and $\rho^{[1]}$ on DC1 and DC2. (a) and (d) 30 dB. (b) and (e) 40 dB. (c) and (f) 50 dB.

structure can improve the unmixing performances and can be treated as a plug-and-play module but may encounter mismatch issues under heavy fine-tuning levels.

4) Can the Proposed Layered Structure Reduce Mutual Coherence of the Library?: Another question this article tends to answer is: can the proposed layered structure reduce mutual coherence of the library? This experiment, which used the Cuprite dataset based on the USGS library, sought to provide an answer to that question. The MUSIC approach [60] is considered for fair comparisons. The proposed layered structure and MUSIC both reduce the 498 atoms in the USGS library to 30 atoms. Fig. 12 shows the mutual coherence [66] obtained from MUSIC and each layer of our proposed layered structure. As shown in Fig. 12, the mutual coherence values returned from MUSIC stand at 1, indicating that MUSIC's goal of mutual coherence reduction was not achieved after the pruning phase. In contrast, the proposed layered structure can gradually decrease the mutual coherence from 1 to 0.9956 as the layer depth increases.

D. Parameters Analysis

The proposed method involves four parameters, i.e., λ, $\rho^{[1]}$, $\tau^{[1]}$, and p. The first parameter measures the degree of importance of sparsity in the abundance matrix, with values varying between $\{1e^{-4}, 5e^{-4}, 1e^{-3}, 5e^{-3}, 1e^{-2}, 5e^{-2}, 1e^{-1}, 5e^{-1}, 1, 5\}$. The second parameter controls the sum-to-one constraint and can be tuned within a range of $\{0, 0.1, 0.5, 1, 5\}$ to achieve the optimal results. The third parameter sets the initial number of iterations and is tuned within a range of $\{5, 10, 20, 30, 40\}$.

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TABLE V
REGULARIZATION PARAMETERS ON THREE DATASETS

	SUnSAL	SUnSAL-TV		CLSUnSAL	S2WSU	MUA			DRSU-TV		TSWCSU-TV		SSLRSU		LGSU			Ours	
	λ	λ	λ_{TV}	λ	λ	λ_C	λ	β	λ	λ_{TV}	λ	λ_{TV}	λ	λ_{LR}	λ_ℓ	λ_g	ψ	λ	$\rho^{[1]}$
DC1	1e-2	5e-3	5e-3	5e-1	1e-2	1e-3	1e-1	1	1e-1	5e-3	1	5e-3	1e-2	5e-2	1e-3	5e-2	7	1e-1	5e-1
	1e-3	5e-3	5e-4	1e-2	1e-3	1e-3	1e-2	1	5e-2	5e-4	1	5e-4	5e-1	1e-2	1e-4	1e-2	5	5e-2	5e-1
	1e-3	1e-3	1e-4	1e-3	1e-4	1e-3	1e-1	1	1e-2	1e-4	1	1e-4	1e-2	5e-3	1e-5	1e-2	3	1e-2	5e-1
DC2	5e-2	5e-2	5e-3	5e-2	1e-2	1e-3	1e-1	1	1	5e-3	1	5e-3	5e-1	1e-3	1e-3	1e-2	3	1	1
	5e-3	5e-3	5e-4	1e-2	1e-3	1e-3	1e-1	1	1e-1	5e-4	1	5e-4	1e-2	5e-2	1e-4	5e-2	3	1e-1	1
	1e-3	1e-3	1e-4	5e-4	1e-3	1e-4	1e-1	1	1e-2	1e-4	1	1e-4	1e-3	1e-2	1e-5	1e-2	3	1e-2	1
Cuprite	1e-3	5e-4	5e-3	1e-2	1e-4	1e-3	1e-3	3	2e-3	1e-4	1e-3	1e-3	3e-3	2e-1	1e-4	1e-2	5	1e-2	1

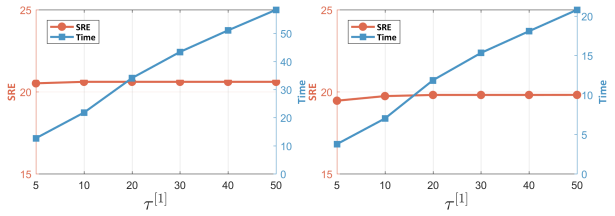


Fig. 14. Unmixing accuracy and consumption regarding $\tau^{[1]}$ on (a) DC1 and (b) DC2.

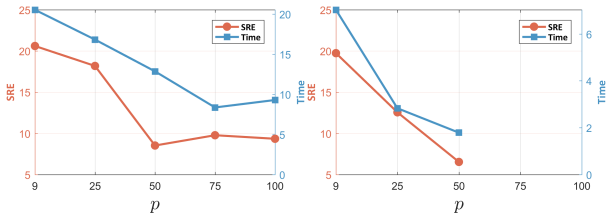


Fig. 15. Unmixing accuracy and consumption regarding p on (a) DC1 and (b) DC2.

The last parameter determines the number of active library atoms measured, which can be estimated using HySime. For the simulated datasets, p is tuned within a range of $\{9, 20, 50, 100\}$. Despite the four parameters mentioned above, we notice that $1/q$ of dictionary atoms is removed from the dictionary at each layer. In this regard, we tune q within a range of $\{2, 3, 4, 5, 6\}$.

1) Unmixing Performances Regarding λ and $\rho^{[1]}$: Fig. 13 shows how the SRE tendency of parameters varies across different noise situations. In DC1, we observe that λ steadily increases and then peaks before dropping quickly. In contrast, adjusting λ in DC2 is much easier due to the proposed method's robustness to changes in λ. In addition, we found that setting $\rho^{[1]}$ to 0.5 leads to more stable unmixing performances compared to fixing it at 0, which indicates that the regularization of the sum-to-one constraint helps improve the unmixing performance. These optimal parameters are summarized in Table V, which equally reports the regularization parameters for most unmixing algorithms on the entire datasets.

2) Unmixing Performances Regarding $\tau^{[1]}$: Fig. 14 shows the impact of initial number of iterations at the first layer to the unmixing performances. The unmixing performances on two DC1 and DC2 show similar behavior that with the increasing of $\tau^{[1]}$, the SRE results slightly go up at the beginning and then remain stable, but the computational time shows an upward trend. This observation indicates that a large $\tau^{[1]}$ does not help boost the unmixing accuracy but requires plenty of computational resources. Thus, $\tau^{[1]}$ is suggested to be fixed at 10.

3) Unmixing Performances Regarding p: Fig. 15 visually shows the impact of different p's to the unmixing performances. As shown in Fig. 15, when p rises, the unmixing accuracy and computational burden drop gradually. A small p indicates a large number of layers and simultaneously demands more computation time. However, since the small p may highly approximate the real distribution of the scene, the unmixing accuracy on the small p will be relatively high.

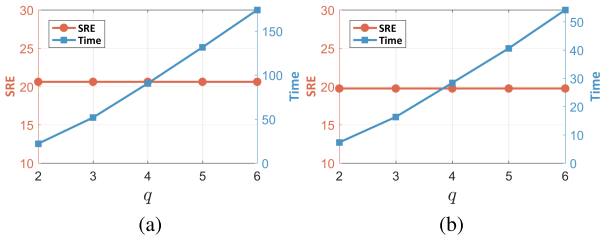


Fig. 16. Unmixing accuracy and consumption regarding q on (a) DC1 and (b) DC2.

4) Unmixing Performances Regarding q: Fig. 16 shows the observation of unmixig performances on different q's. It can be observed from Fig. 16 that the unmixing results are robustness to the selection of q. However, a large q increases the computational burden as only $1/q$ of libraries is believed to be inactive and required to be removed. In the proposed LSU, q is fixed at 2, meaning that the half of libraries will be removed at each layer.

5) Unmixing Performances Regarding the Different Number of Active Libraries Versus Total Libraries: To observe the impact of different number of active library on the unmixing performances. This experiment experiments using simulated hyperspectral datasets generated by the synthesis toolbox[13] because this toolbox can generate the datasets according to different settings. The simulated datasets had a size of 150×150 pixels, with the number of endmembers ranging from 5 to 50 in intervals of 5, and a fixed noise level of 30 dB. The spectral library used for unmixing the datasets consisted of 30 signatures and 99 bands, randomly selected from the JSC library. The angular separation between any pair of spectral signatures was at least 4.44°.

Fig. 17 shows the SRE values obtained from various algorithms on different dataset scenarios. Fig. 17 clearly shows that as the number of endmembers approaches the total number

[13]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Imagery_Synthesis_tools_for_MATLAB

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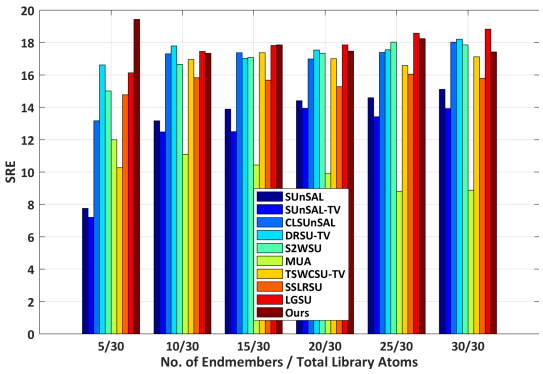


Fig. 17. Tendency of SRE values with respect to varying numbers of endmembers in the total library atoms.

of library atoms, most algorithms exhibit an increasing trend in SRE values. These results indicate that when the number of endmembers matches the total number of library atoms, it reduces the presence of unnecessary atoms that introduce noise. In addition, it ensures the inclusion of necessary atoms for accurately representing the mixture, enabling the methods to effectively estimate the contributions of each active atom without introducing any additional noise or ambiguity.

IV. Conclusion

This article discusses sparse regression unmixing, a technique used to estimate the abundance fractions of mixed pixels in an image by identifying the optimal subset of atoms from a given library. While previous algorithms focused solely on either abundance estimation or library fine-tuning, this article introduces a layered sparse regression algorithm, called LSU, that can perform both tasks in multilayers. At each layer, the algorithm estimates the row-sparsity-promoting abundance matrix in the inner loop according to the fine-tuned library and identifies the active library atoms in the outer loop based on the activeness measured sparse rows of the learned abundance matrix. Using ADMM to solve the optimization problem at each layer and stopping when certain criteria are met, this approach is able to efficiently exploit the unmixing performance of sparse regression and fine-tune libraries based on sparse row measurements. The effectiveness and efficiency of the proposed method are demonstrated through experimental results on both simulated and real hyperspectral datasets.

Appendix

This section details the optimization procedures of the proposed LSU. For simplicity, we only formulate the optimization procedures of LSU at the ℓ layer. We first reformulate (6) by the following constrained formulation:

$$\begin{aligned}\min_{X^{[\ell]},V_1^{[\ell]},V_2^{[\ell]},V_3^{[\ell]},V_4^{[\ell]}} \ & \frac{1}{2}\left\|Y-V_1^{[\ell]}\right\|_F^2+\lambda\left\|W^{[\ell]}\otimes V_2^{[\ell]}\right\|_1 \\ & +\frac{\rho^{[\ell]}}{2}\left\|\mathbf{1}_{\delta^{[\ell]}}^T V_3^{[\ell]}-\mathbf{1}_N^T\right\|_F^2+\iota_{\mathbb{R}_+}(V_4) \\ \text{s.t. } & V_1^{[\ell]}=A^{[\ell]}X^{[\ell]},\quad V_2^{[\ell]}=X^{[\ell]} \\ & V_3^{[\ell]}=X^{[\ell]},\quad V_4^{[\ell]}=X^{[\ell]}. \end{aligned} \tag{15}$$

Then, (15) can be expressed in an augmented Lagrangian form

$$\begin{aligned}&\mathcal{L}_\mu\left(X^{[\ell]},V_1^{[\ell]},V_2^{[\ell]},V_3^{[\ell]},V_4^{[\ell]},D_1^{[\ell]},D_2^{[\ell]},D_3^{[\ell]},D_4^{[\ell]}\right) \\ &=\frac{1}{2}\left\|Y-V_1^{[\ell]}\right\|_F^2+\frac{\mu}{2}\left\|A^{[\ell]}X^{[\ell]}-V_1^{[\ell]}-D_1^{[\ell]}\right\|_F^2 \\ &\quad+\lambda\left\|W_1^{[\ell]}\otimes V_2^{[\ell]}\right\|_1+\frac{\mu}{2}\left\|X^{[\ell]}-V_2^{[\ell]}-D_2^{[\ell]}\right\|_F^2 \\ &\quad+\frac{\rho^{[\ell]}}{2}\left\|\mathbf{1}_{\delta^{[\ell]}}^T V_3^{[\ell]}-\mathbf{1}_N^T\right\|_F^2+\frac{\mu}{2}\left\|X^{[\ell]}-V_3^{[\ell]}-D_3^{[\ell]}\right\|_F^2 \\ &\quad+\iota_{\mathbb{R}_+}\left(V_4^{[\ell]}\right)+\frac{\mu}{2}\left\|X^{[\ell]}-V_4^{[\ell]}-D_4^{[\ell]}\right\|_F^2 \end{aligned} \tag{16}$$

where $\mu>0$ denotes a penalty parameter and $D_1^{[\ell]}$, $D_2^{[\ell]}$, $D_3^{[\ell]}$, and $D_4^{[\ell]}$ denote scaled Lagrangian multipliers.

Update $X^{[\ell]}$: The optimization subproblem of $X^{[\ell]}$ is

$$\begin{aligned}\arg\min_{X^{[\ell]}} \ & \frac{\mu}{2}\left\|A^{[\ell]}X^{[\ell]}-V_1^{[\ell]}-D_1^{[\ell]}\right\|_F^2 \\ & +\frac{\mu}{2}\left\|X^{[\ell]}-V_2^{[\ell]}-D_2^{[\ell]}\right\|_F^2 \\ & +\frac{\mu}{2}\left\|X^{[\ell]}-V_3^{[\ell]}-D_3^{[\ell]}\right\|_F^2 \\ & +\frac{\mu}{2}\left\|X^{[\ell]}-V_4^{[\ell]}-D_4^{[\ell]}\right\|_F^2 \end{aligned} \tag{17}$$

where its closed-form solution can be obtained by

$$X^{[\ell]}=F^{-1}\times\left(\left(A^{[\ell]}\right)^T\psi_1^{[\ell]}+\psi_2^{[\ell]}+\psi_3^{[\ell]}+\psi_4^{[\ell]}\right) \tag{18}$$

where $F=(A^{[\ell]})^TA^{[\ell]}+3I$, $\psi_1^{[\ell]}=V_1^{[\ell]}+D_1^{[\ell]}$, $\psi_2^{[\ell]}=V_2^{[\ell]}+D_2^{[\ell]}$, $\psi_3^{[\ell]}=V_3^{[\ell]}+D_3^{[\ell]}$, and $\psi_4^{[\ell]}=V_4^{[\ell]}+D_4^{[\ell]}$

Update $V_1^{[\ell]}$: The optimization subproblem of $V_1^{[\ell]}$ is

$$\arg\min_{V_1^{[\ell]}}\frac{1}{2}\left\|Y-V_1^{[\ell]}\right\|_F^2+\frac{\mu}{2}\left\|A^{[\ell]}X^{[\ell]}-V_1^{[\ell]}-D_1^{[\ell]}\right\|_F^2. \tag{19}$$

The closed-from solution of (19) can be obtained by

$$V_1^{[\ell]}=\frac{1}{1+\mu}\left[Y+\mu\left(A^{[\ell]}X^{[\ell]}-D_1^{[\ell]}\right)\right]. \tag{20}$$

Update $V_2^{[\ell]}$: The optimization subproblem of $V_2^{[\ell]}$ is

$$\arg\min_{V_2^{[\ell]}}\lambda\left\|W^{[\ell]}\otimes V_2^{[\ell]}\right\|_1+\frac{\mu}{2}\left\|X^{[\ell]}-V_2^{[\ell]}-D_2^{[\ell]}\right\|_F^2. \tag{21}$$

The solution of (21) can be obtained by

$$V_2^{[\ell]}\leftarrow\text{SoftTh}\left(X^{[\ell]}-D_2^{[\ell]},W^{[\ell]}\lambda/\mu\right) \tag{22}$$

where $\text{SoftTh}(\circ,\star)=\text{sign}(\circ)\max(|\circ|-\star,0)$.

Update $V_3^{[\ell]}$: The optimization subproblem of $V_3^{[\ell]}$ is

$$\arg\min_{V_3}\frac{\rho^{[\ell]}}{2}\left\|\mathbf{1}_{\delta^{[\ell]}}^TV_3^{[\ell]}-\mathbf{1}_N^T\right\|_F^2+\frac{\mu}{2}\left\|X^{[\ell]}-V_3^{[\ell]}-D_3^{[\ell]}\right\|_F^2. \tag{23}$$

The solution of (23) can be obtained by

$$\begin{aligned}V_3^{[\ell]}\leftarrow&\left(\rho^{[\ell]}\mathbf{1}_{\delta^{[\ell]}\times\delta^{[\ell]}}+\mu I\right)^{-1} \\ &\times\left(\rho^{[\ell]}\mathbf{1}_{\delta^{[\ell]}\times N}+\mu\left(X^{[\ell]}-D_3^{[\ell]}\right)\right). \end{aligned} \tag{24}$$

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Update $V_4^{[\ell]}$*:* The optimization subproblem of $V_4^{[\ell]}$ is

$$\arg\min_{V_4^{[\ell]}} \iota_{\mathbb{R}_+}\left(V_4^{[\ell]}\right) + \frac{\mu}{2}\left\|X^{[\ell]} - V_4^{[\ell]} - D_4^{[\ell]}\right\|_F^2. \tag{25}$$

The solution of (25) can be obtained by a nonnegative projection operator

$$V_4^{[\ell]} \leftarrow \max\left(X^{[\ell]} - D_4^{[\ell]}, 0\right). \tag{26}$$

Update Multipliers: The four Lagrangian multipliers are updated by the following formulas:

$$\begin{cases} D_1^{[\ell]} = D_1^{[\ell]} + V_1^{[\ell]} - A^{[\ell]}X^{[\ell]} \\ D_2^{[\ell]} = D_2^{[\ell]} + V_2^{[\ell]} - X^{[\ell]} \\ D_3^{[\ell]} = D_3^{[\ell]} + V_3^{[\ell]} - X^{[\ell]} \\ D_4^{[\ell]} = D_4^{[\ell]} + V_4^{[\ell]} - X^{[\ell]}. \end{cases} \tag{27}$$

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